

A-FREE MEASURES AND APPLICATIONS

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ABSTRACT. It is the note of the lecture by De Philippis at Zhejiang University. Most of the materials are from the lecture note [4]. However, the lecture did not strictly follow that note. We intend to present a comprehensive course record and provide some personal understanding and solutions for exercises.

1. SOME EXAMPLES AND MOTIVATIONS

1.1. Some definitions and examples.

Definition 1.1. We define

$$\mathcal{A} := \sum_{|\alpha|=k} A_\alpha \partial^\alpha$$

as a k -homogeneous linear differential operator, where $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$ is a multi-index with

$$|\alpha| = \sum_{i=1}^d \alpha_i, \quad \partial^\alpha = \prod_{i=1}^d \partial_i^{\alpha_i},$$

and $A_\alpha \in L(V, W)$ is a linear transform from V to W .

Indeed, we see that

$$\mathcal{A} : C^\infty(\Omega, V) \rightarrow C^\infty(\Omega, W)$$

is well-defined, where $\Omega \subset \mathbb{R}^d$ is a domain and V, W are two finite dimensional vector spaces.

We can extend $\mathcal{A}$ in Definition 1.1 to all distributions $T \in \mathcal{D}'(\Omega, V)$. To make it clear, we define V -valued distributions.

Definition 1.2. T is a V -valued distribution, or $T \in \mathcal{D}'(\Omega, V)$ if it is a linear and continuous operator on $C_0^\infty(\Omega, V^*)$, where V^* is the dual space of V .

For $T \in \mathcal{D}'(\Omega, V)$ and $\mathcal{A}$ as in Definition 1.1, we define

$$\mathcal{A}T : C_0^\infty(\Omega, W^*) \rightarrow C_0^\infty(\Omega, V^*)$$

such that

$$\langle \mathcal{A}T, \varphi \rangle = \langle T, \mathcal{A}^* \varphi \rangle \quad \text{for any } \varphi \in C_0^\infty(\Omega, W^*),$$

where

$$\mathcal{A}^* := \sum_{|\alpha|=k} (-1)^{|\alpha|} A_\alpha^* \partial^\alpha$$

is the dual operator of $\mathcal{A}$. Note that $A_\alpha \in L(W^*, V^*)$ is the dual operator of A_α .

Here we have some examples.

Example 1.3. The Laplace operator

$$\Delta : C_0^\infty(\Omega, \mathbb{R}^\ell) \rightarrow C_0^\infty(\Omega, \mathbb{R}^\ell)$$

is given as

$$\Delta u^i := \sum_{j=1}^d \partial_j^2 u^i \quad \text{for any } i \in \mathbb{Z} \cap [1, \ell].$$

Example 1.4. The divergence operator

$$\operatorname{div} : C_0^\infty(\Omega, \mathbb{R}^d) \rightarrow C_0^\infty(\Omega, \mathbb{R})$$

is given as

$$\operatorname{div} u := \sum_{j=1}^d \partial_j u^j.$$

We can extend div as the operator from $C_0^\infty(\Omega, \mathbb{R}^\ell \oplus \mathbb{R}^d)$ to $C_0^\infty(\Omega, \mathbb{R}^\ell)$. Indeed,

$$(\operatorname{div} T)_j := \sum_{i=1}^d \partial_i T_j^i$$

for any $j \in \mathbb{Z} \cap [1, \ell]$. Intuitively, we view it as a row-wise divergence. Here, T represents the matrix with ℓ -rows and d -columns.

Example 1.5. The curl operator

$$\operatorname{curl} : C_0^\infty(\Omega, \mathbb{R}^d) \rightarrow C_0^\infty(\Omega, \mathbb{R}^{\frac{d(d-1)}{2}})$$

is defined as

$$(\operatorname{curl} T)_{ij} := \partial_i T^j - \partial_j T^i.$$

Here, we note that $\operatorname{curl} T$ is an anti-symmetric matrix, so it has the value in $\mathbb{R}^{\frac{d(d-1)}{2}}$. Analogous to div , we can extend it to row-wise sense. We define

$$\operatorname{curl} : C_0^\infty(\Omega, \mathbb{R}^\ell \oplus \mathbb{R}^d) \rightarrow C_0^\infty(\Omega, \mathbb{R}^\ell \oplus \mathbb{R}^{\frac{d(d-1)}{2}})$$

as

$$(\operatorname{curl} T)_{ij}^k := \partial_i T^{jk} - \partial_j T^{ik}$$

for any $k \in \mathbb{Z} \cap [1, \ell]$.

Example 1.6. The exterior differential operator

$$d : C_0^\infty(\Omega, \wedge^\ell(\mathbb{R}^d)) \rightarrow C_0^\infty(\Omega, \wedge^{\ell+1}(\mathbb{R}^d)).$$

Let $\mathcal{A}$ be as in Definition 1.1, we have the following definition.

Definition 1.7. A distribution T is $\mathcal{A}$ -free, if $\mathcal{A}T = 0$.

1.2. Motivations. With the definitions in the previous subsection, we present some motivations of the study for $\mathcal{A}$ -free elements.

Motivation 1. In many situations, $\mathcal{A}u = 0$ (u is $\mathcal{A}$ -free) if and only if there is a differential operator B such that $u = Bv$ for some v .

In the following examples, we assume that all functions and maps are sufficiently smooth for simplicity.

Example 1.8. $\operatorname{curl} u = 0$ if and only if there is v such that $u = Dv$.

Example 1.9. Let $\Omega = B_1$. A ℓ -form is closed ($dw = 0$) if and only if there is $\eta \in C_0^\infty(B_1, \wedge^{\ell-1}(\mathbb{R}^d))$ such that $\omega = d\eta$.

Example 1.10. $v = \frac{(\nabla u) + (\nabla u)^T}{2}$ if and only if $\operatorname{curl} \operatorname{curl} v = 0$. Here, we note that

$$(\operatorname{curl} \operatorname{curl} v)_{jh} := \sum_i (\partial_{ih}^2 v_{ij} + \partial_{ij}^2 v_{ih} - \partial_{jh}^2 v_{ii} - \partial_{ii}^2 v_{jh}).$$

Example 1.11. We can express the $W^{1,p}$ as

$$W^{1,p}(\Omega, \mathbb{R}^\ell) = \{u \in L^p(\Omega, \mathbb{R}^\ell \oplus \mathbb{R}^d) : \operatorname{curl} u = 0\}.$$

Note that $\operatorname{curl} u = 0$ means that there is $v \in L^p$ such that $u = Dv$.

Example 1.12. We have the definition of BV functions as

$$BV(\Omega, \mathbb{R}^\ell) := \{u \in L^1(\Omega, \mathbb{R}^\ell) : \nabla u \text{ is a Radon measure}\}.$$

Then, it is equivalent to

$$\{\mu \in \mathcal{M}(\Omega, \mathbb{R}^\ell \oplus \mathbb{R}^d) : \text{curl } \mu = 0\}.$$

Example 1.13. Define the space of functions with bounded deformations as

$$BD(\Omega, \mathbb{R}^d) := \{u \in L^1(\Omega, \mathbb{R}^d) : (\nabla u) + (\nabla u)^T \text{ is a Radon measure}\}.$$

It reduces to

$$\{\mu \in \mathcal{M}(\Omega, \mathbb{R}^\ell \oplus \mathbb{R}^d) : \text{curl curl } \mu = 0\}.$$

Remark 1.14. Definitions in Example 1.11, 1.12, and 1.13 are not rigorous. They are here to give some intuitions about motivation.

Motivation 2. We can write all non-linear PDEs as a linear PDE system + (non-linear) pointwise constraints.

Example 1.15. Consider the standard elliptic equation

$$\text{div}(A(\nabla u)) = 0 \tag{1.1}$$

with $A_i : \mathbb{R}^d \rightarrow \mathbb{R}^d$. Precisely, for $u \in C^\infty(\Omega, \mathbb{R}^d)$, $A(\nabla u) = \sum_{i=1}^n A_i(\partial_i u)$, where $A_i : \mathbb{R}^d \rightarrow \mathbb{R}^d$. Then, we can interpret it to the equivalent system

$$\text{curl } v = 0, \quad \text{div } w = 0, \quad w = Av. \tag{1.2}$$

Note that here the equivalence means that (1.1) has a solution if and only if (1.2) has a solution. Later, we will give more background on the equation (1.1).

Example 1.16. Though $\Delta u = 0$ is not a non-linear equation, it is also beneficial to consider its translation to the system. Indeed, it corresponds to

$$\text{curl } v = 0, \quad \text{div } w = 0, \quad w = v.$$

Example 1.17. Let $u : \Omega \rightarrow \mathbb{R}^d$ and $F : \mathbb{R}^d \rightarrow \mathbb{R}^d$. Consider the conservation law $\partial_t u - \text{div}(F(u)) = 0$. It is equivalent to

$$\partial_t u - \text{div } w = 0, \quad w = F(u).$$

Motivation 3. (Calculus of variations) We study the variational problem

$$\min_{\substack{u \in \text{function space} \\ + \text{boundary conditions}}} \int_{\Omega} F(\nabla u) dx, \tag{1.3}$$

where $u : \Omega \rightarrow \mathbb{R}^\ell$ and $F : \mathbb{R}^\ell \oplus \mathbb{R}^d \rightarrow \mathbb{R}$.

For problem (1.3), we examine optimality conditions as follows.

Euler-Lagrange equations. Assume that $u : \Omega \rightarrow \mathbb{R}^\ell$ is a minimizer. For $\varphi \in C_0^\infty(\Omega, \mathbb{R}^\ell)$, we consider $u_\varepsilon := u + \varepsilon \varphi$ as the perturbation of u and have that

$$\int_{\Omega} F(\nabla u) dx \leq \int_{\Omega} F(\nabla u_\varepsilon) dx := g(\varepsilon)$$

for any $\varepsilon > 0$. Then

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} g = 0. \tag{1.4}$$

Through simple expansion,

$$\int_{\Omega} F(\nabla u_\varepsilon) dx = \int_{\Omega} F(\nabla u) dx + \varepsilon \int_{\Omega} \nabla F(\nabla u) \cdot \nabla \varphi dx + O(\varepsilon^2).$$

As a result, (1.4) implies that for any $C_0^\infty(\Omega, \mathbb{R}^\ell)$,

$$\int_{\Omega} \nabla F(\nabla u) \cdot \nabla \varphi dx = 0.$$

In the distributional sense, we have that

$$\operatorname{div}(A(\nabla u)) = 0, \quad A = \nabla F,$$

which is the equation in Example (1.15).

Inner perturbation. Let $X \in C_0^\infty(\Omega, \mathbb{R}^d)$. The map $\Phi_\varepsilon(x) : x + \varepsilon X(x)$ is a diffeomorphism of Ω if $\varepsilon > 0$ is sufficiently small. Let $u_\varepsilon = u \circ \Phi_\varepsilon$. We let

$$g(\varepsilon) := \int_{\Omega} F(\nabla u_\varepsilon) dx.$$

Similarly, (1.4) still holds. To calculate the expansion of $g(\varepsilon)$ at 0, we need

$$\nabla \Phi_\varepsilon(x) = \operatorname{Id} + \varepsilon \nabla X(x), \quad (\nabla \Phi_\varepsilon(x))^{-1} = \operatorname{Id} - \varepsilon \nabla X(x) + O(\varepsilon^2).$$

As a result,

$$\det \nabla \Phi_\varepsilon(x) = 1 + \varepsilon \operatorname{tr}(\nabla X) + O(\varepsilon^2) = 1 + \varepsilon \operatorname{div} X + O(\varepsilon^2).$$

Combining all the calculations above, we have that

$$\begin{aligned} g(\varepsilon) &= \int F((\nabla \Phi_\varepsilon^{-1})^T (\nabla u \circ \Phi_\varepsilon^{-1})) dx \\ &= \int F((\nabla \Phi_\varepsilon^{-1})^T (\Phi_\varepsilon(x)) (\nabla u(x))) \det(\Phi_\varepsilon(x)) dx \\ &= \int F(\nabla u) dx + \varepsilon \int (F(\nabla u) \operatorname{div} X - \langle \nabla F(\nabla u), (\nabla X)^T \nabla u \rangle) dx + O(\varepsilon^2), \end{aligned}$$

where for the second equality, we use the change of variables. Then,

$$\int (F(\nabla u) \operatorname{div} X - \langle \nabla F(\nabla u), (\nabla X)^T \nabla u \rangle) dx = 0$$

for any $X \in C_0^\infty(\Omega, \mathbb{R}^d)$. As a result,

$$\int_{\Omega} (F(\nabla u) \operatorname{Id} - \nabla F(\nabla u) \otimes \nabla u) (\nabla X) dx = 0.$$

Define

$$T = T_u := F(\nabla u) \operatorname{Id} - \nabla F(\nabla u) \otimes \nabla u$$

as the stress tensor, we see that it implies that in the sense of distribution, $\operatorname{div} T = 0$.

Assume that we have a sequence of local minimizers $\{u_j\}$ of the problem (1.3) with uniform bounded energy, i.e.

$$\sup_{j \in \mathbb{Z}_+} \int_{\Omega} F(\nabla u_j) dx < +\infty. \quad (1.5)$$

Then, up to a subsequence,

$$\mu_j := F(\nabla u_j) dx \rightharpoonup^* \mu \in \mathcal{M}(\Omega)$$

in the sense of Radon measures. We now assume that the stress tensor satisfies the pointwise estimate

$$|T_u| \lesssim |F(\nabla u)| + |\nabla F(\nabla u)| |\nabla u| \lesssim |F(\nabla u)|. \quad (1.6)$$

Note that for $F(\xi) := |\xi|^p$ with $p > 1$, such a property is satisfied. On the other hand, (1.5) and (1.6) imply that up to a subsequence,

$$T_{u_j} \rightharpoonup^* T \text{ in } \mathcal{M}(\Omega, \mathbb{R}^{d \times d}).$$

Since $|T_{u_j}| \lesssim \mu_j$, we have

$$|T|(A) \leq C \mu(A) \quad \text{for any } A \subset \Omega.$$

Then, it follows from the Radon-Nikodym theorem that there exists $P \in L^\infty(\mu)$ such that $T = P\mu$. Roughly, we have

$$P(x) = \int (F(y) \operatorname{Id} - \nabla F(y) \otimes y) d\nu_x(y),$$

where $\nu_x(y)$ is some measure.

We recall famous results as in [8], giving an example for the above analysis.

Theorem 1.18. Assume that $\{u_j\} \subset H^1(\Omega, \mathcal{N})$ is a sequence of stationary harmonic maps with

$$\int_{\Omega} |\nabla u_j|^2 \leq \Lambda,$$

where $\mathcal{N}$ is a smooth and compact manifold. We have that up to a subsequence,

$$|\nabla u_j|^2 dx \rightharpoonup^* |\nabla u|^2 dx + \nu \text{ in } \mathcal{M}(\Omega)$$

with ν being the defect measure. Moreover, it supports on a $(d-2)$ -rectifiable set Σ .

After class, I asked Professor De Philippis if such a notation is similar to the notions of stationary measures as in [9]. He replied that the framework of his can include such a discussion.

1.3. Non-convex variational problem. Let $Q := [0, 1]^2$ be the two-dimensional unit cube. Consider the variational problem

$$\inf_{u \in W_0^{1,4}(Q)} \int_Q ((u_x^2 - 1)^2 + u_y^2) dx dy := I(u). \quad (1.7)$$

For such a variational problem, we have the following two properties.

Proposition 1.19. *There is no $u \in H_0^1(Q)$ such that $I(u) = 0$.*

Proof. Note that if $I(u) = 0$, we must have that $u_x = \pm 1$ and $u_y = 0$. As a result, we see that u is independent of the variable y and satisfies

$$\partial_x u(x, y) = \chi_E(x) - \chi_{[0,1] \setminus E}(x),$$

where $E \subset [0, 1]$. Obviously, u is not vanishing on ∂Q , and then $u \notin W_0^{1,4}(Q)$. $\square$

Proposition 1.20. (1.7) achieves the inf value 0.

Proof. We construct $\varphi_j : [0, 1] \rightarrow [0, 1]$ such that $\text{supp } \varphi_j \subset [j^{-1}, 1 - j^{-1}]$, $\varphi_j \equiv 1$ in $[2j^{-1}, 1 - 2j^{-1}]$ and $|\varphi_j'| \leq 10j^{-1}$. Define $\psi_k : [0, 1] \rightarrow [0, 1]$ such that for any $x \in [\frac{i}{2k}, \frac{i+2}{2k}]$ with i being even, we have

$$\psi_k(x) = \begin{cases} x - \frac{i}{2k} & \text{if } x \in [\frac{i}{2k}, \frac{i+1}{2k}], \\ -x + \frac{i+1}{2k} & \text{if } x \in [\frac{i+1}{2k}, \frac{i+2}{2k}]. \end{cases}$$

Let $u_{jk} = \varphi_j(y)\psi_k(x)$. Then, it follows that

$$0 \leq I(u_{jk}) \leq \int_{\{\varphi_j \neq 1\}} 1 dx dy + \frac{1}{2k} \int_Q |\varphi_j'|^2 dx dy.$$

For any $\varepsilon > 0$, we first choose $j \gg 1$ such that the first term is less than $\frac{\varepsilon}{2}$. Then, the whole term is less than ε if we further determine that the second term is less than $\frac{\varepsilon}{2}$. Then we find a sequence u_{j,k_j} such that $I(u_{j,k_j}) \rightarrow 0$ as $j \rightarrow +\infty$. $\square$

In the above example, we see that for a functional

$$\int_{\Omega} F(\nabla u) dx,$$

there are no solutions to $\{F(\nabla u) = 0\}$ but there are solutions u_j such that

$$\lim_{j \rightarrow +\infty} \text{dist}(\nabla u_j, \{F = 0\}) = 0.$$

Exercise 1.21. Let $F(\eta) = \text{dist}^2(\eta, \{\text{Id}, -\text{Id}\})$ where $\eta \in \mathbb{R}^2 \otimes \mathbb{R}^2$ and Id is the identity matrix. Prove that

$$\inf_{u \in H_0^1(Q, \mathbb{R}^2)} \int_Q F(\nabla u) > 0.$$

In the class, the professor mentioned that there is no eliminatory solution, and one needs to use advanced theory to solve Exercise 1.21. We delay the proof in the proceeding context of this note.

2. DIFFERENTIAL INCLUSION

2.1. Motivation. In Section 1.3, the Exercise 1.21 motivates us to consider the following problem

$$\inf_{u \in H_0^1(\Omega, \mathbb{R})} \int_{\Omega} \text{dist}^2(\nabla u, \{a, b\}) dx dy, \quad (2.1)$$

where $\Omega \subset \mathbb{R}^d$ is a bound domain and $a, b \in \mathbb{R}^d$. To consider (2.1), we consider the following two questions.

The first is the exact solutions, that is

$$\nabla u(x) \in \{a, b\} := K. \quad (2.2)$$

Through the notation curl, we interpret it into the form

$$v \in \{a, b\}, \quad \text{curl } v = 0.$$

The second is the approximation solutions. Namely, it refers to $\{u_j\}$ such that

$$\text{dist}(\nabla u_j, \{a, b\}) \rightarrow 0$$

in some sense (maybe in a.e. or L^p). In other words, we have

$$\text{dist}(v_j, \{a, b\}) \rightarrow 0, \quad \text{curl } v_j = 0.$$

From (2.2), we have that

$$\nabla u(x) = a\chi_E(x) + b\chi_{E^c}(x).$$

Then,

$$u(x) = a \cdot x + h((b - a) \cdot x), \quad (2.3)$$

where the derivative h' takes values in $\{0, 1\}$. The proof of (2.3) is due to the translation and change of variable such that we can consider the most trivial case $a = (1, 0, \dots, 0)$ and $b = (-1, 0, \dots, 0)$. If we consider the situation that $a = (1, 0)$ and $b = (-1, 0)$ with $\Omega = Q = [0, 1]^2$, we can construct u_j just as those in Proposition 1.19 and obtain u_j such that

$$\text{dist}(\nabla u_j, \{a, b\}) \rightarrow 0 \text{ in } L^2(Q).$$

Such a property implies that for any admissible function, the value of the functional in (2.1) is positive, but the variational problem is zero.

The problem above naturally lets us study the following differential inclusion problem. Let $K \subset \mathbb{R}^\ell \oplus \mathbb{R}^d$ be a compact set. We consider the problem

$$\nabla u \in K.$$

Equivalently, it is to say

$$v \in K, \quad \text{curl } v = 0.$$

There are still two questions.

- Classify the exact solutions of $\nabla u \in K$.
- Find the approximation solutions $\{u_j\}$ such that

$$\lim_{j \rightarrow +\infty} \text{dist}(\nabla u_j, K) = 0$$

in a.e. or L^p some sense.

Let us first give some examples of the exact solutions.

Example 2.1. If $K = \mathbb{O}(d)$ is a collection of the orthonormal matrices, then u is the rigidity of $\mathbb{R}^d$.

Proof. Note that $\|\nabla u\|_\infty \leq 1$. Also, it follows from the chain rule that

$$\left| \frac{d}{dt} u(\gamma(t)) \right| = |\gamma'(t)|$$

for any curve γ . Then u conserves the metric and is a rigidity of $\mathbb{R}^d$. □

Example 2.2. If

$$K = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} : a, b \in \mathbb{R} \right\},$$

then the exact solutions correspond to the collection of all holomorphic functions since we can easily verify the Cauchy-Riemann equation.

Next, we present a theorem that concerns a more generalized form as the differential inclusion problem (2.1).

Theorem 2.3 (Ball-James [3]). *Let $\Omega \subset \mathbb{R}^d$ be bounded domain and let $A, B \in \mathbb{R}^\ell \otimes \mathbb{R}^d$ be $(\ell \times d)$ -matrices. Then, the following properties hold.*

- (1) *If $\text{rank}(A - B) = 1$, then the exact solutions in $W^{1,\infty}(\Omega, \mathbb{R}^\ell)$ are in the form*

$$u(x) = Ax + \xi h(x \cdot m),$$

where $B - A = \xi \otimes m$. In addition, we have a sequence approximation solutions $\{u_j\}$ such that

$$\text{dist}(\nabla u_j, \{A, B\}) \rightarrow 0$$

a.e. and there exists $\varepsilon > 0$, depending on A, B such that

$$\min \left\{ \int_{\Omega} |\nabla u_j - A|, \int_{\Omega} |\nabla u_j - B| \right\} > \varepsilon.$$

- (2) *If $\text{rank}(A - B) \geq 2$, then we have results as follows.*

- (a) *If $u \in W^{1,\infty}(\Omega, \mathbb{R}^\ell)$ satisfies the differential inclusion*

$$\nabla u(x) \in \{A, B\} \quad \text{for a.e. } x \in \Omega,$$

then either $\nabla u \equiv A$ or $\nabla u \equiv B$.

- (b) *Let $\{u_j\} \subset W^{1,\infty}(\Omega, \mathbb{R}^\ell)$ be a uniformly norm-bounded sequence of maps such that*

$$\text{dist}(\nabla u_j, \{A, B\}) \rightarrow 0 \quad \text{in the sense of Radon measure.}$$

Then, up to a subsequence, either

$$\int_{\Omega} |\nabla u_j - A| \rightarrow 0 \quad \text{or} \quad \int_{\Omega} |\nabla u_j - B| \rightarrow 0$$

as $j \rightarrow +\infty$.

For the case that $\text{rank}(A - B) = 1$, the analysis is similar to the problem (2.1). When $\text{rank}(A - B) \geq 2$, we only show the classification of exact solutions for some intuitions. Later, we will give more general results covering the properties in the above theorem.

Proof of the exact solutions for $\text{rank}(A - B) \geq 2$. We see that $\nabla u(x) \in \{A, B\}$ a.e. in Ω . As a result,

$$\nabla u(x) = A\chi_E(x) + B\chi_{E^c}(x).$$

By letting $v = u - Bx$, we have

$$\nabla v(x) = (A - B)\chi_E(x) := C\chi_E(x).$$

Since $\text{curl}(Dv) = 0$, it follows that for any $k \in \mathbb{Z} \cap [1, \ell]$,

$$0 = \partial_i(c_j^k \chi_E) - \partial_j(c_i^k \chi_E) = c_j^k \partial_i \chi_E - c_i^k \partial_j \chi_E.$$

Noting that $\text{rank } C \geq 2$, we see that there must be two $k \in \mathbb{Z} \cap [1, \ell]$ such that $(c_1^k, \dots, c_d^k)$ with these k s are linear independent and paralleled to $\nabla \chi_E$. Then $\partial \chi_E \equiv 0$, proving that E must be Ω or $\emptyset$ in a.e. sense. $\square$

2.2. A more general setting. Previously, we consider the derivative inclusion problem. Here, we have a generalized form, changing the operator ∇ to $\mathcal{A}$, the homogeneous derivative operator. We now consider the problem

$$v \in K \subset W, \quad \mathcal{A}v = 0. \quad (2.4)$$

Example 2.4. The problem

$$\frac{\nabla u + (\nabla u)^T}{2} \in K$$

is equivalent to

$$\operatorname{curl} \operatorname{curl} v = 0, \quad v \in K.$$

A special case is the 2-states problem. We first have the convention and the notation as in Definition 1.1 that

$$K = \{\lambda_1, \lambda_2\} \subset W, \quad \mathcal{A} = \sum_{|\alpha|=k} A_\alpha \partial^\alpha.$$

The issue is understanding the exact and approximate solutions of the problem (2.4). Theorem 2.3 motivates us to first find exact solutions in the form as

$$v(x) = \lambda_1 + (\lambda_2 - \lambda_1) \chi_I(x \cdot \xi),$$

where $I \subset \mathbb{R}$. Here, the logic is that if we have already eliminated the rank 1 case in Theorem 2.3, then the remaining case is like those for rank ≥ 2 . Through simple calculations, we have

$$\begin{aligned} \mathcal{A}v &= (\lambda_2 - \lambda_1) \sum_{|\alpha|=k} A_\alpha \partial^\alpha \chi_I(x \cdot \xi) \\ &= \left((\lambda_2 - \lambda_1) \sum_{|\alpha|=k} A_\alpha \xi^\alpha \right) (\partial^k \chi_I(x \cdot \xi)), \quad \xi^\alpha = \prod_{i=1}^d \xi_i^{\alpha_i}. \end{aligned}$$

As a result, we see that $\mathcal{A}v = 0$ if and only if

$$\lambda_2 - \lambda_1 \in \bigcup_{|\xi|=1} \ker \mathbb{A}(\xi),$$

where

$$\mathbb{A}(\xi) := \sum_{|\alpha|=k} A_\alpha \xi^\alpha \in L(V, W). \quad (2.5)$$

We now have the following definition.

Definition 2.5 (Symbols and wave cones). Let $\mathcal{A}$ be as in Definition 1.1. We define its symbol as the operator (2.5) and the wave cone as

$$\Lambda_{\mathcal{A}} := \bigcup_{|\xi|=1} \ker \mathbb{A}(\xi).$$

Recall the notation of the Fourier transform

$$\mathcal{F}(f)(\xi) = \widehat{f}(\xi) := \int_{\mathbb{R}^d} f(x) \exp(2\pi i x \cdot \xi) dx.$$

With the help of Fourier transform, we have

$$\mathcal{F}(\mathcal{A}f)(\xi) = (2\pi i)^k \mathbb{A}(\xi) \widehat{f}(\xi). \quad (2.6)$$

Here, the notion symbol is analogous to that in micro-local analysis.

Definition 2.6 (Ellipticity). A differential operator

$$\mathcal{B} : \mathcal{D}'(\Omega, V) \rightarrow \mathcal{D}'(\Omega, W)$$

as in Definition 1.1 is elliptic if $\mathbb{B}(\xi)$ is injective for any $\xi \neq 0$. For such a case, $\dim V$ must be less or equal to $\dim W$.

Remark 2.7. The intuition of the definition for the ellipticity is that in (2.6), if we want to recover the original elements f from $\mathcal{A}f$ through the Fourier transform, we need to multiply both sides with the reverse of $\mathbb{A}(\xi)$.

Remark 2.8. The ellipticity in the above definition is also coherent with the classical definition for equations. Recall that

$$a_{ij}\partial_i\partial_j u = 0$$

is an elliptic equation if

$$\sum_{i,j=1}^d a_{ij}\xi_i\xi_j > 0$$

for any $\xi \neq 0$.

Lemma 2.9. $\lambda \notin \Lambda_{\mathcal{A}}$ if and only if the operator

$$\mathcal{B}_{\lambda} : \mathcal{D}'(\Omega, \mathbb{R}) \rightarrow \mathcal{D}'(\Omega, W)$$

defined as

$$\mathcal{B}_{\lambda}(u) := \mathcal{A}(\lambda u) \quad \text{for any } u \in \mathcal{D}'(\Omega, \mathbb{R})$$

is elliptic.

Proof. Assume that $\mathcal{B}_{\lambda}$ is not elliptic, then for some $\xi \neq 0$, the symbol

$$\mathbb{B}_{\lambda}(\xi) = \sum_{|\alpha|=k} A_{\alpha} \lambda \xi^{\alpha}$$

is not injective. As a result, $\mathbb{B}_{\lambda}(\xi) = 0$, implying that $\lambda \in \Lambda_{\mathcal{A}}$. On the other hand, if $\lambda \in \Lambda_{\mathcal{A}}$, the reverse arguments directly yield the result. $\square$

Example 2.10. We give some examples as follows.

(1) If $\mathcal{A} = \Delta$, then

$$\ker \mathbb{A}(\xi) = \{0\} \quad \text{for any } \xi \neq 0, \quad \Lambda_{\mathcal{A}} = \{0\}.$$

(2) If $\mathcal{A} = \text{curl}$, then

$$\ker \mathbb{A}(\xi) = \{a \otimes \xi\} \quad \text{for any } \xi \neq 0, \quad \Lambda_{\mathcal{A}} = \{a \otimes b : a \in \mathbb{R}^{\ell}, b \in \mathbb{R}^d\}.$$

(3) If $\mathcal{A} = \text{curl curl}$, then

$$\ker \mathbb{A}(\xi) = \{a \odot \xi\} \quad \text{for any } \xi \neq 0, \quad \Lambda_{\mathcal{A}} = \{a \odot b : a, b \in \mathbb{R}^d\}.$$

where $a \odot b = a \otimes b + b \otimes a$.

(4) If $\mathcal{A} = \text{div}$, then

$$\ker \mathbb{A}(\xi) = \{M \in \mathbb{R}^d \oplus \mathbb{R}^d : M\xi = 0\}, \quad \Lambda_{\mathcal{A}} = \{M \in \mathbb{R}^d \oplus \mathbb{R}^d : \det M = 0\}.$$

Proof. If $\mathcal{A} = \Delta$, we see that

$$\Delta : C_0^{\infty}(\Omega, \mathbb{R}) \rightarrow C_0^{\infty}(\Omega, \mathbb{R}).$$

Then $\mathbb{A}(\xi) = |\xi|^2$ is a multiplier on $\mathbb{R}$. As a result, for any λ with $\lambda|\xi|^2 = 0$, it follows that $\lambda = 0$ since $\xi \neq 0$.

If $\mathcal{A} = \text{curl}$ with $\xi \neq 0$, then for $T \in \ker \mathbb{A}(\xi)$, we have

$$\xi_i T_j^k - \xi_j T_i^k = 0$$

for any $i, j \in \mathbb{Z} \cap [1, d]$ and $k \in \mathbb{Z} \cap [1, \ell]$. Consequently, there must be some $a \in \mathbb{R}^{\ell}$ such that $T = a \otimes \xi$.

Assume that $\mathcal{A} = \text{curl curl}$. Up to a rotation, without loss of generality, we assume that $\xi = (1, 0, \dots, 0)$. For $v \in \ker \mathbb{A}(\xi)$, we deduce that

$$\sum_{i=1}^d (\xi_i \xi_h v_{ij} + \xi_i \xi_j v_{ih} - \xi_j \xi_h v_{ii} - \xi_i^2 v_{jh}) = 0$$

for any $j, h \in \mathbb{Z} \cap [1, d]$. Note that then,

$$\xi_h v_{1j} + \xi_j v_{1h} - \xi_j \xi_h v_{11} - v_{jh} = 0 \quad (2.7)$$

for any $j, h \in \mathbb{Z} \cap [1, d]$. If $h = 1$, we have

$$v_{1j} = v_{j1} \quad \text{for any } j \in \mathbb{Z} \cap [1, d].$$

If $j \neq 1$ and $h \neq 1$, then (2.7) implies that $v_{jh} = 0$. Letting $a_j = v_{1j}$ for $j \neq 1$ and $a_1 = \frac{1}{2}v_{11}$, it yields that $v = a \odot \xi$.

Since $\operatorname{div} M = \sum_{j=1}^d \partial_j M_j$, we see that if $\xi \neq 0$ with $\xi \in \ker \mathbb{A}(\xi)$, then

$$\sum_{j=1}^d \xi_j M_{ij} = 0$$

for any $i \in \mathbb{Z} \cap [1, d]$. As a result, $M\xi = 0$. It follows that

$$\Lambda_{\operatorname{div}} = \{M \in \mathbb{R}^d \oplus \mathbb{R}^d : \det M = 0\},$$

completing the proof. $\square$

Next, we give a generalized form of Theorem 1.6.

Theorem 2.11 (De Philippis-Palmieri-Rindler [5]). *Let $\Omega \subset \mathbb{R}^d$ be bounded domain and $\mathcal{A}$ be as in Definition 1.1. Suppose that $\lambda_1, \lambda_2 \in \mathbb{R}^\ell$ with $\lambda_1 - \lambda_2 \notin \Lambda_{\mathcal{A}}$. Then, the following properties hold.*

- (1) *If $u \in L^\infty(\Omega, \mathbb{R}^\ell)$ is $\mathcal{A}$ -free and*

$$u(x) \in \{\lambda_1, \lambda_2\} \quad \text{for a.e. } x \in \Omega,$$

then either $u = \lambda_1$ or $u = \lambda_2$ a.e. in Ω .

- (2) *Let $p \in (1, +\infty)$ and let $\{u_j\} \subset L^p(\Omega, \mathbb{R}^\ell)$ be a uniformly norm-bounded sequence of $\mathcal{A}$ -free functions. Assume that*

$$\lim_{j \rightarrow +\infty} \int_{\Omega} \operatorname{dist}^p(u_j(x), \{\lambda_1, \lambda_2\}) dx = 0. \quad (2.8)$$

Then, up to a subsequence, either

$$\int_{\Omega} |u_j(x) - \lambda_1|^p dx \rightarrow 0 \quad \text{or} \quad \int_{\Omega} |u_j(x) - \lambda_2|^p dx \rightarrow 0 \quad (2.9)$$

as $j \rightarrow +\infty$.

Remark 2.12. When $\Omega = \mathbb{R}^d$, the first property also holds.

Remark 2.13. We see that this theorem implies the second point of Theorem 2.3.

To prove the above theorem, we need the following technical lemma.

Lemma 2.14. *Let $\mathcal{A}$ be a differential operator of order k and let $\varphi \in C_0^\infty(\Omega)$, then there exists a differential operator $\mathcal{C}$ of order at most $k-1$ (but may not homogeneous)*

$$\mathcal{C} = \sum_{|\alpha| \leq k-1} C_\alpha \partial^\alpha$$

such that for all $u \in L^p(\Omega, \mathbb{R}^m)$ there exists a compactly supported $v \in L^p(\Omega, \mathbb{R}^N)$, $N = N(m, k, d)$ such that

$$[\mathcal{A}, \varphi]u := \mathcal{A}(\varphi u) - \varphi \mathcal{A}(u) = \mathcal{C}v.$$

Moreover for any $p \in [1, +\infty)$,

$$\|v\|_{L^p(\Omega, \mathbb{R}^N)} \leq C \|u\|_{L^p(\Omega, \mathbb{R}^m)}$$

where C depends only on k, d , and φ .

Proof. We see that $[\mathcal{A}, \varphi](u)$ is a sum of forms as

$$\partial^\beta \varphi \partial^\gamma u$$

with $|\gamma| \leq k-1$. We let $k=2$ and $\gamma, \beta = 1$ for simplicity. For the general case, one can prove with similar arguments. As a result,

$$\partial_1 \varphi \partial_1 u = \partial_1(\partial_1 \varphi u) - \partial_{11} \varphi u.$$

Let $v = (v_1, v_2) := (\partial_1 \varphi u, -\partial_{11}^2 \varphi u)$. We have $\mathcal{C}v = \partial_1 v_1 + v_2$, and the result follows. $\square$

Before we present the proof of Theorem 2.11, we first apply it to solve Exercise 1.21.

Proof of the result in Exercise 1.21. Assume that $\{u_j\}$ is a sequence of minimizing sequence. It follows from Theorem 2.11 that $\nabla u_j \rightarrow \text{Id}$ or $\nabla u_j \rightarrow -\text{Id}$ in L^2 . It directly gives the contradiction to the minimizing property of $\{u_j\}$. $\square$

Proof of Theorem 2.11(1). Without loss of generality, we assume that $\lambda_1 = 0$ and $\lambda_2 = \lambda \notin \Lambda_{\mathcal{A}}$. If not, we consider $u - \lambda_1$. Since $u(x) \in \{0, \lambda\}$ for a.e. $x \in \Omega$, we have that $u(x) = \lambda \chi_E(x)$, where $E \subset \Omega$. Note that

$$\mathcal{A}u = 0, \quad u = \lambda \chi_E(x).$$

As a result, $\mathcal{B}_\lambda(\chi_E) = 0$. If $\Omega = \mathbb{R}^d$, then it follows from Fourier transform that

$$\mathbb{B}_\lambda(\xi) \widehat{\chi}_E(\xi) = 0.$$

Using Lemma 2.9, we see that unless $\xi \neq 0$, $\mathbb{B}_\lambda(\xi) \neq 0$. Consequently, $\text{supp } \widehat{\chi}_E \subset \{0\}$, implying that $E = \Omega$ or $\emptyset$ in the sense of almost everywhere. For the case that Ω is a bounded domain, we apply the cut-off function φ . Indeed, we have

$$[\mathcal{B}_\lambda, \varphi](\chi_E) = \mathcal{B}_\lambda(\varphi u) - \varphi \mathcal{B}_\lambda(\chi_E)$$

and

$$\mathcal{B}_\lambda(\varphi \chi_E) = \varphi \mathcal{B}_\lambda(\chi_E) + [\mathcal{B}_\lambda, \varphi](\chi_E) = [\mathcal{B}_\lambda, \varphi](\chi_E),$$

where $\varphi \in C_0^\infty(\mathbb{R}^d)$. With the help of Lemma 2.14, there is $v \in L^2$ such that

$$\|[\mathcal{B}_\lambda, \varphi](\chi_E)\|_{H^{-(k-1)}} \leq C \|v\|_{L^2} \leq C \|\chi_E\|_{L^2}. \quad (2.10)$$

Through Fourier transform, we deduce that

$$\widehat{\varphi \chi_E}(\xi) = \frac{\mathbb{B}_\lambda(\xi)}{1 + |\mathbb{B}_\lambda(\xi)|^2} \cdot \widehat{f}(\xi) + \frac{\widehat{\varphi \chi_E}(\xi)}{1 + |\mathbb{B}_\lambda(\xi)|^2}, \quad (2.11)$$

where up to a constant,

$$\widehat{f}(\xi) = \mathbb{B}_\lambda(\xi) \widehat{\varphi \chi_E}(\xi).$$

Note that

$$\frac{\mathbb{B}_\lambda(\xi)}{1 + |\mathbb{B}_\lambda(\xi)|^2} \sim \frac{1}{|\xi|^k}, \quad \frac{1}{1 + |\mathbb{B}_\lambda(\xi)|^2} \sim \frac{1}{|\xi|^{2k}}.$$

This, together with (2.10) and (2.11), implies that

$$|\xi| \widehat{\varphi \chi_E}(\xi) \in L^2.$$

Then $\varphi \chi_E \in H^1$. By the arbitrariness of φ , we see that E must be Ω or empty set in a.e. sense. $\square$

The proceeding task is proving Theorem 2.11(2). Let us first present related remarks.

Remark 2.15. Given (2.8), we see that

$$\sup_{j \in \mathbb{Z}_+} \|u_j\|_{L^p(\Omega)} < +\infty.$$

Up to a subsequence, $u_j \rightharpoonup u$ weakly in $L^p(\Omega)$. Since $\mathcal{A}u_j = 0$ is in the distribution sense, we have $\mathcal{A}u = 0$. To this end, Theorem 2.11(2) follows directly if we can verify the strong convergence of u_j . If $u_j \rightarrow u$ strongly in $L^p(\Omega)$, then up to a subsequence, such a convergence is also in a.e. sense, showing that $u \in K$ a.e. in Ω . As a result, we can apply Theorem 2.11(1) to complete the proof.

Note that the weak convergence does not imply the a.e. convergence. Let us consider an example. Let $f : \mathbb{T} \rightarrow \mathbb{R}$ such that $f \equiv 0$ in $[0, \frac{1}{2}]$ and $f \equiv 1$ in $[\frac{1}{2}, 1]$. Then, $f_j = f(\frac{\cdot}{2j}) \rightharpoonup \frac{1}{2}$ weakly in $L^2(\mathbb{R})$. Intuitively, weak convergence takes the average of oscillation.

Proof of Theorem 2.11(2). As shown in the above remark, it remains to prove that up to a subsequence, u_j is convergent in $L^p(\Omega)$. We first consider the case $p = 2$ as a special case and then deal with general $p \in (1, +\infty)$ with the help of results in harmonic analysis.

Analogous to the proof of Theorem 2.11(1), we still assume that $\lambda_1 = 0$ and $\lambda_2 = \lambda \neq \Lambda_{\mathcal{A}}$. Let

$$u_j := \lambda \chi_{E_j} + r_j \quad \text{and} \quad E_j := \{x \in \Omega : |u_j - \lambda| < \varepsilon_j\},$$

where $\{\varepsilon_j\}$ is a positive sequence to be determined such that

$$\lim_{j \rightarrow +\infty} \varepsilon_j = 0.$$

Since $\mathcal{A}u_j = 0$, we have

$$\mathcal{B}_\lambda(\chi_{E_j}) = \mathcal{A}(\lambda \chi_{E_j}) = -\mathcal{A}(r_j), \quad (2.12)$$

where $\mathcal{B}_\lambda$ is as in Lemma 2.9. Note that

$$\int_{\Omega} r_j^2 = \int_{E_j} r_j^2 + \int_{E_j^c} r_j^2 = \int_{E_j} |r_j - \lambda \chi_{E_j}|^2 + \int_{E_j^c} u_j^2. \quad (2.13)$$

By the definition of E_j ,

$$\int_{E_j} |r_j - \lambda \chi_{E_j}|^2 \leq \varepsilon_j^2 \mathcal{L}^d(\Omega). \quad (2.14)$$

On the other hand,

$$\begin{aligned} \int_{E_j^c} u_j^2 &= \left(\int_{E_j^c \cap \{|u_j| \leq \varepsilon_j\}} + \int_{E_j^c \cap \{|u_j| > \varepsilon_j\}} \right) u_j^2 \\ &\leq \varepsilon_j^2 \mathcal{L}^d(\Omega) + \int_{\{\text{dist}(u_j, \{0, \lambda\}) \geq \varepsilon_j\}} u_j^2. \end{aligned}$$

By the property that

$$|u_j| \leq |\lambda| + \text{dist}(u_j, \{0, \lambda\}),$$

it follows that

$$\int_{E_j^c} u_j^2 \leq C \left(|\lambda| \mathcal{L}^d(\{x \in \Omega : \text{dist}(u_j(x), \{0, \lambda\}) \geq \varepsilon_j\}) + \int_{\Omega} \text{dist}^2(u_j, \{0, \lambda\}) \right). \quad (2.15)$$

Using Chebyshev inequality,

$$\mathcal{L}^d(\{x \in \Omega : \text{dist}(u_j(x), \{0, \lambda\}) \geq \varepsilon_j\}) \leq \frac{1}{\varepsilon_j^2} \int_{\Omega} \text{dist}^2(u_j, \{0, \lambda\}). \quad (2.16)$$

Letting ε_j be as

$$\varepsilon_j^2 = \left(\int_{\Omega} \text{dist}^2(u_j, \{0, \lambda\}) \right)^{\frac{1}{2}}.$$

This, together with (2.8), (2.15) and (2.16), implies that

$$\int_{E_j^c} u_j^2 \rightarrow 0$$

as $j \rightarrow +\infty$. Combining (2.13) and (2.14),

$$\lim_{j \rightarrow +\infty} \int_{\Omega} u_j^2 = 0. \quad (2.17)$$

Note that with almost the same arguments, (2.17) is valid for $p \neq 2$, namely,

$$\lim_{j \rightarrow +\infty} \int_{\Omega} |u_j|^p = 0. \quad (2.18)$$

Back to the equation (2.12), we consider the cut-off function $\varphi \in C_0^\infty(\Omega)$ and obtain

$$\mathcal{B}_\lambda(\varphi\chi_{E_j}) = \varphi\mathcal{A}(r_j) + [\mathcal{B}_\lambda, \varphi](\chi_{E_j}) = \mathcal{A}(\varphi r_j) + [\mathcal{A}, \varphi](r_j) + [\mathcal{B}_\lambda, \varphi](\chi_{E_j}). \quad (2.19)$$

We remark the necessity to apply the cut-off function is that (2.12) holds only in Ω and may not be true on $\mathbb{R}^d$ due to the potential value-jump on the boundary. By Lemma 2.14, there is $v_j \in L^2(\Omega)$ such that

$$[\mathcal{A}, \varphi](r_j) + [\mathcal{B}_\lambda, \varphi](\chi_{E_j}) = \mathcal{C}(v_j)$$

and for any $j \in \mathbb{Z}_+$ and $p \in (1, +\infty)$,

$$\|v_j\|_{L^p} \leq C(\|u_j\|_{L^p} + \|r_j\|_{L^p}) \leq C. \quad (2.20)$$

Note that $\mathcal{C}$ is a differential operator with order no more than $k-1$. Taking Fourier transform for both sides of (2.19) and letting $\mathbb{B}_\lambda(\xi)$, $\mathbb{A}(\xi)$, and $\mathbb{C}(\xi)$ be corresponding symbols of $\mathcal{B}_\lambda$, $\mathcal{A}$, and $\mathcal{C}$, it follows that

$$\mathbb{B}_\lambda(\xi)\widehat{\varphi\chi_{E_j}}(\xi) = \mathbb{A}(\xi)\widehat{\varphi r_j}(\xi) + \mathbb{C}(\xi)\widehat{v_j}(\xi),$$

where we omit some constants. As a result, we have the identity

$$\widehat{\varphi\chi_{E_j}}(\xi) = \frac{\mathbb{B}_\lambda(\xi) \cdot \mathbb{A}(\xi)\widehat{\varphi r_j}(\xi)}{1 + |\mathbb{B}_\lambda(\xi)|^2} + \frac{\mathbb{B}_\lambda(\xi) \cdot \mathbb{A}(\xi)\widehat{v_j}(\xi)}{1 + |\mathbb{B}_\lambda(\xi)|^2} + \frac{\widehat{\varphi\chi_{E_j}}(\xi)}{1 + |\mathbb{B}_\lambda(\xi)|^2}, \quad (2.21)$$

where we note that here $\cdot$ means the inner product in W . Using Lemma 2.9, since $\lambda \notin \Lambda_{\mathcal{A}}$, we have

$$|\mathbb{B}_\lambda(\xi)| \geq c_0|\xi|^k.$$

Noting that

$$\|\mathbb{A}(\xi)\| \sim |\xi|^k, \quad \|\mathbb{C}(\xi)\| \sim |\xi|^{k-1},$$

As a result, for any $R > 1$, we deduce from (2.20) and (2.21) that

$$\begin{aligned} \int_{\mathbb{R}^d \setminus B_R} |\widehat{\varphi\chi_{E_j}}(\xi)|^2 d\xi &\leq C \int_{\mathbb{R}^d \setminus B_R} \|\widehat{\varphi r_j}\|^2 dx + C \int_{\mathbb{R}^d \setminus B_R} \frac{|\widehat{v_j}(\xi)|^2}{(1 + |\xi|)} d\xi + C \int_{\mathbb{R}^d \setminus B_R} \frac{|\widehat{\varphi\chi_{E_j}}(\xi)|^2}{1 + |\xi|^{2k}} d\xi \\ &\leq C(\|r_j\|_{L^2} + R^{-1}). \end{aligned}$$

Given (2.17), Corollary A.3 and the arbitrariness of $\varphi \in C_0^\infty(\Omega)$, we see that $\{\chi_j\}$ is precompact, completing the proof.

For the case that $p \neq 2$, we sketch the proof. With the help of (2.20) and (2.21) we have

$$\varphi\chi_{E_j} = \mathcal{F}^{-1}(m_0(\xi)\mathcal{F}(r_j)) + \mathcal{F}(m_1(\xi)\mathcal{F}(w_j)), \quad (2.22)$$

where

$$\sup_{j \in \mathbb{Z}_+} \|w_j\|_{L^p} \leq C, \quad (2.23)$$

and the multipliers m_0 and m_1 satisfy

$$|\partial^\beta m_0(\xi)| \leq C_\beta |\xi|^{-\beta}, \quad |\partial^\beta (\sqrt{1 + |\xi|^2} m_1(\xi))| \leq C_\beta |\xi|^{-\beta}$$

for any multi-index $\beta = (\beta_1, \dots, \beta_d)$. Using (2.18) and Theorem A.4, we see that

$$\|\mathcal{F}^{-1}(m_0(\xi)\mathcal{F}(r_j))\|_{L^p} \rightarrow 0$$

as $j \rightarrow +\infty$. On the other hand,

$$\mathcal{F}(m_1(\xi)\mathcal{F}(w_j)) = \tilde{K} * (\mathcal{F}^{-1}(1 + |\xi|^2)^{-\frac{1}{2}} * w_j).$$

The convolution with $\tilde{K}$ gives a strong (p, p) bound. Moreover, the convolution with $\mathcal{F}^{-1}(1 + |\xi|^2)^{-\frac{1}{2}}$ represents a compact operator in L^p . We first note that

$$\mathcal{F}^{-1}(1 + |\xi|^2)^{-\frac{1}{2}} \sim \begin{cases} \frac{1}{|x|^{d-1}} & \text{if } |x| \leq 1, \\ e^{-|x|} & \text{if } |x| \geq 1. \end{cases}$$

It implies that $\mathcal{F}^{-1}(1 + |\xi|^2)^{-\frac{1}{2}} \in L^1$. Using Theorem A.1, we can prove $\{\mathcal{F}(m_1(\xi)\mathcal{F}(w_j))\}_j$ is precompact. We remark that up to a cut-off function, one only needs to verify the first and third properties of Theorem A.1, which are obvious with the help of Yang's inequality for

convolutions. Combining (2.22) and (2.23), we obtain that up to a subsequence, χ_{E_j} converges strongly in $L^p(\Omega)$. $\square$

To close this subsection, we recall the following Vitali lemma concerning the relations between weak and strong convergence in L^p . The proof is simple, and we omit it here.

Lemma 2.16 (Vitali). *Let Ω be a bounded domain. A sequence $\{u_j\} \subset L^p(\Omega)$ satisfies $\sup_j \|u_j\|_{L^p} \leq +\infty$. u_j converges strongly to u in $L^p(\Omega)$ if and only if the following properties hold.*

(1) *There is no concentration, i.e.*

$$\lim_{\mathcal{L}^d(E) \rightarrow 0^+} \sup_{j \in \mathbb{Z}_+} \int_E |u_j|^p = 0.$$

(2) *There is no oscillation, namely, for any $\varepsilon > 0$*

$$\mathcal{L}^d(\{|u_j - u| > \varepsilon\}) \rightarrow 0^+$$

as $j \rightarrow +\infty$.

3. $\mathcal{A}$ -FREE MEASURES

3.1. Some Motivations. Note that in Theorem 2.11, we assume that $p \in (1, +\infty)$. That is because the convergence with $p = 1$ is complex. Let $\{u_j\} \subset L^1(\Omega)$ be such that

$$\sup_{j \in \mathbb{Z}_+} \|u_j\|_{L^1(\Omega)} < +\infty.$$

Then, there is $\mu \in \mathcal{M}(\Omega, \mathbb{R})$ such that up to a subsequence, $u_j dx \rightharpoonup^* \mu$ in $\mathcal{M}(\Omega, \mathbb{R})$. For example $u_j := j\chi_{[0, j^{-1}]}$ $\rightharpoonup^* \delta_0$. Also, since $\mathcal{A}(u_j) = 0$ in the sense of distribution, it follows from the convergence $u_j \rightharpoonup^* \mu$ that $\mathcal{A}(\mu) = 0$. Now, we present some intentions as follows.

- Understanding the concentration.
- Understanding the structure of function spaces like $BV(\Omega)$ and $BD(\Omega)$.
- Understanding the structure of Lipschitz functions and metric spaces.

The first two points are easy to follow. We recall Rademacher's theorem, giving intuitions for the third point above.

Theorem 3.1 (Rademacher's theorem). *Every Lipschitz map $f : \mathbb{R}^d \rightarrow \mathbb{R}^m$ is differentiable $\mathcal{L}^d$ -a.e. in $\mathbb{R}^d$.*

A natural question is as follows.

Question 3.2. What are measures with Rademacher's theorem? μ is a desired measure if the Lipschitz map is differentiable μ -a.e. in $\mathbb{R}^d$.

We mainly focus on Question 3.2. For $\mu = f dx$, the Rademacher's theorem is true. When $\mu = \delta_0$ we have the counterexample $f(x) = |x|$. For $\mu = \mathcal{H}^k \llcorner \Sigma$ with $\mathcal{H}^k(\Sigma) < +\infty$, it is also false since we have the example $f(x) = \text{dist}(x, \Sigma)$.

Regarding Question 3.2, we have the following conjecture.

Conjecture 3.3. *If μ is a Radon measure with Rademacher's theorem being true, then $\mu \ll \mathcal{L}^d$.*

Concerning the conjecture, we consider the reverse side and use the contradiction arguments. We need the following result.

Theorem 3.4 (Alberti-Marchese, [1]). *Assume that $\nu \perp \mathcal{L}^d$ is a singular measure of $\mathcal{L}^d$ and satisfies the Rademacher's theorem, then there are $\{\mu_i\}_i^d \subset \mathcal{M}(\Omega, \mathbb{R}^d)$ such that the following properties hold.*

- (1) $\nu \ll \mu_i$ for any $i = 1, 2, \dots, d$.
- (2) $\text{div}(\mu_i) = 0$ for any $i = 1, 2, \dots, d$.
- (3) For ν -a.e. x ,

$$\text{span} \left\{ \frac{d\mu_1}{d|\mu_1|}(x), \dots, \frac{d\mu_d}{d|\mu_d|}(x) \right\} = \mathbb{R}^d.$$

Remark 3.5. Later, we will apply this theorem to solve Conjecture 3.3

3.2. Main theorem. Let $\mu \in \mathcal{M}(\Omega, V)$ be a vector-valued radon measure. By the Radon-Nikodym theorem, we have the decomposition

$$\mu = \mu^a + \mu^s = g d\mathcal{L}^d + \frac{d\mu}{d|\mu|} |\mu|^s.$$

Note that $g \in L^1(\Omega, V)$ and $|\mu|$ is the total variation of μ . In particular, we call

$$\frac{d\mu}{d|\mu|} \in L^\infty_{|\mu|}(\Omega, V)$$

as the polar vector of μ . We have the definition

$$\frac{d\mu}{d|\mu|}(x) = \lim_{r \rightarrow 0^+} \frac{\mu(B_r(x))}{|\mu|(B_r(x))}$$

and that the limit exists $|\mu|$ -a.e. in $\mathbb{R}^d$.

Theorem 3.6 (De Philippis-Palmieri-Rindler [5]). *Let $\Omega \subset \mathbb{R}^d$ be an open set and $\mathcal{A}$ be a differential operator as in Definition 1.1. Assume that μ is an $\mathcal{A}$ -free Radon measure on Ω . Then*

$$\frac{d\mu}{d|\mu|}(x) \in \Lambda_{\mathcal{A}} \quad \text{for } |\mu|^s\text{-a.e. } x \in \Omega.$$

Before we give the details of the proof, we first present some applications.

Given the wave cone calculations in Example (2.10), we see that

$$\Lambda_{\text{curl}} = \{a \otimes \xi : a \neq 0, \xi \neq 0\}.$$

It implies that for $u \in BV(\Omega)$,

$$\text{rank} \left(\frac{d(\nabla u)}{d|\nabla u|} \right) = 1$$

for $|\nabla u|^s$ -a.e. $x \in \Omega$. Also, for $\mathcal{A} = \text{curl curl}$, we see that

$$\Lambda_{\mathcal{A}} = \{a \odot \xi : a \neq 0, \xi \neq 0\}.$$

As a result,

$$\frac{d(Eu)}{d|Eu|} = a \odot \xi$$

for $|Eu|^s$ -a.e. $x \in \Omega$, where $Eu = \frac{\nabla u + (\nabla u)^T}{2}$.

Next, we will apply results for Λ_{div} to solve Conjecture 3.3.

Proof of Conjecture 3.3. Note that for

$$\bar{\mu} := (\mu_1, \mu_2, \dots, \mu_d)^T \in \mathcal{M}(\Omega, \mathbb{R}^d \oplus \mathbb{R}^d),$$

we have

$$\text{div } \bar{\mu} := (\text{div } \mu_1, \text{div } \mu_2, \dots, \text{div } \mu_d)^T \in \mathcal{M}(\mathbb{R}^d, \mathbb{R}^d \oplus \mathbb{R}^d).$$

Since

$$\Lambda_{\text{div}} = \{M \in \mathbb{R}^d \oplus \mathbb{R}^d : \det M = 0\},$$

we see from Theorem 3.6 that

$$\text{rank} \left(\frac{d\bar{\mu}}{d|\bar{\mu}|} \right) \leq d - 1$$

for $\sum_{i=1}^d |\mu_i|^s$ -a.e. x . On the other hand, Theorem 3.4 tells us that for the singular ν ,

$$\text{rank} \left(\frac{d\bar{\mu}}{d|\bar{\mu}|} \right) = d$$

for ν -a.e. x . Since $\nu \ll \nu_i$ for any $i \in \mathbb{Z} \cap [1, d]$, we see that

$$\nu^s \ll \sum_{i=1}^d |\mu_i|^s,$$

which is a contradiction. □

To show Theorem 3.6, we present the blow-up analysis of measures.

For a vector-valued Radon measure $\mu \in \mathcal{M}(\Omega, V)$, $x \in \Omega$ and $R \in (0, \text{dist}(x, \partial\Omega))$, we define

$$\mu_{x,R}(A) := \frac{\mu(x + RA)}{|\mu|(B_R(x))},$$

for any measurable set $A \subset B_1$. As a result, we see that $|\mu_{x,R}|(B_1) \leq 1$. Up to a subsequence $R_j \rightarrow 0^+$, we can take the limit of μ_{x,R_j} in the sense of Radon measures. Precisely, we have the following theorem.

Theorem 3.7 (Preiss [10]). *Assume that μ is a vector-valued Radon measure. Then for $|\mu|$ -a.e. x , there exists $R_j \rightarrow 0^+$ such that $\mu_{x,R_j} \rightharpoonup^* \nu$ in B_1 and the following properties hold.*

- (1) $|\nu|$ has the uniform bound $|\nu|(B_1) \leq 1$.
- (2) ν is non-trivial, i.e. $\nu \llcorner B_{\frac{1}{2}} \neq 0$.
- (3) There is a positive measure σ such that

$$\mu = \frac{d\mu}{d|\mu|} \sigma.$$

Moreover, as $j \rightarrow +\infty$,

$$|\mu|_{x,R_j} \rightharpoonup^* \sigma.$$

Remark 3.8. Indeed, $\sigma = |\nu|$. ν depends on the choice of R_j . Call that ν is a tangent measure of μ at the point x . We denote the collection of all tangent measures of μ at x as $\text{Tan}(\mu, x)$.

Proof of Theorem 3.6. Define

$$E := \left\{ x \in \Omega : \frac{d\mu}{d|\mu|}(x) \in \Lambda_{\mathcal{A}} \right\}.$$

Assume that the result is false, namely, $|\mu|^s(E) > 0$. We have $x_0 \in E$ and $R_j \rightarrow 0^+$ such that $\nu \in \text{Tan}(\mu, x_0)$. Without loss of generality, we assume that $x_0 = 0$ and ν satisfies properties in Theorem 3.7 with $\sigma = |\nu|$. Note that

$$\nu = \frac{d\mu}{d|\mu|}(0)\sigma, \quad \lambda := \frac{d\mu}{d|\mu|}(0) \notin \Lambda_{\mathcal{A}}.$$

Note that for $|\mu|^s$ -a.e. x ,

$$\lim_{j \rightarrow +\infty} \frac{|\mu|^s(B_{R_j}(x))}{|\mu|(B_{R_j}(x))} = 1, \quad \lim_{j \rightarrow +\infty} \int_{B_{R_j}(x)} \left| \frac{d\mu}{d|\mu|}(y) - \frac{d\mu}{d|\mu|}(x) \right| d|\mu|(y) = 0. \quad (3.1)$$

Consequently, we assume that the point 0 also satisfies (3.1). Then, σ is also a weak* limit of $(|\mu|^s)_{0,R_j}$, i.e. $\sigma \in \text{Tan}(|\mu|^s, 0)$.

Recall that $\mathcal{A}(\mu_j) = 0$. Since it is in a distributional sense, we have $\mathcal{A}(\lambda\sigma) = 0$. As in Lemma 2.9, we see that $\mathcal{B}_\lambda$ is elliptic and $\mathcal{B}_\lambda(\sigma) = 0$. Using generalized Weyl's lemma, $\sigma \ll \mathcal{L}^d$.¹

However, there is still no contradiction since there is an example of singular measure $\tilde{\sigma}$ such that for $\tilde{\sigma}$ -a.e. x , $\text{Tan}(\tilde{\sigma}, x) = \{\mathcal{L}^d\}$. We need further properties to conclude the proof. Recall that

$$\sigma_j = (|\mu|^s)_{0,R_j} \rightharpoonup^* \sigma$$

as $j \rightarrow +\infty$. By (3.1), we have

$$\lim_{j \rightarrow +\infty} |\mu_j - \lambda\sigma_j|(B_1) = 0, \quad (3.2)$$

¹Weyl's lemma implies that if a function is vanishing under an elliptic operator, then it must be smooth. We can generalize it to Radon measures. Indeed, if a Radon measure becomes zero after an elliptic operator, then it must be absolutely continuous to $\mathcal{L}^d$.

where $\mu_j = \mu_{0,R_j}$. Indeed, we have

$$\begin{aligned} |\mu_j - \lambda\sigma_j| &\leq \frac{|\lambda(|\mu|^s)_{0,R_j} - \mu_{0,R_j}|}{|\mu|^s(B_{R_j})} \\ &\leq \frac{|\lambda|\mu|^s - \mu^s|(B_{R_j})}{|\mu|^s(B_{R_j})} + \frac{|\mu|^a(B_{R_j})}{|\mu|^s(B_{R_j})} \\ &= \int_{B_{R_j}} \left| \frac{d\mu}{d|\mu|}(0) - \frac{d\mu}{d|\mu|}(x) \right| d|\mu|^s(x) + \frac{|\mu|^a(B_{R_j})}{|\mu|^s(B_{R_j})} \rightarrow 0^+. \end{aligned}$$

We claim that

$$\lim_{j \rightarrow +\infty} |\sigma_j - \sigma|(B_1) = 0. \quad (3.3)$$

If such a claim is true, σ is singular of zero. That is because if $\sigma \ll \mathcal{L}^d$, we can find a subset E_j such that

$$\sigma_j(B_1 \setminus E_j) = 0, \quad \mathcal{L}^d(E_j) = 0.$$

As a result,

$$\sigma_j(B_1) = \sigma_j(E_j) = \sigma(E_j) + (\sigma_j - \sigma)(E_j) = (\sigma_j - \sigma)(E_j) \rightarrow 0.$$

The third equality follows from $\sigma \ll \mathcal{L}^d$, and the limit is a consequence of (3.3). Then, σ is trivial in B_1 , a contradiction to Theorem 3.7 that ν is non-trivial in $B_{\frac{1}{2}}$. It remains to show (3.3). The proof is similar to the proof of Theorem 2.11(2). Note that

$$\mathcal{B}_\lambda(\sigma_j) = \mathcal{A}(\lambda\sigma_j) = \mathcal{A}(\mu_j - \lambda\sigma_j).$$

Let $r_j := \mu_j - \lambda\sigma_j$. We put the cut-off function $\varphi \in C_0^\infty(\Omega, \mathbb{R}_{\geq 0})$ and obtain that

$$\mathcal{B}_\lambda(\varphi\sigma_j) = \mathcal{A}(\varphi r_j) + [\mathcal{B}_\lambda, \varphi](\sigma_j) + [\mathcal{A}, \varphi](r_j).$$

Using Lemma 2.14, we find w_j such that

$$\mathcal{B}_\lambda(\varphi\sigma_j) = \mathcal{A}(\varphi r_j) + \mathcal{C}(w_j),$$

where $\mathcal{C}$ has order at most $(k-1)$ and

$$|w_j|(B_1) \leq C(|\sigma_j|(B_1) + |r_j|(B_1)) \leq C.$$

Analogous to (2.21), we have

$$\widehat{\varphi\sigma_j}(\xi) = \frac{\mathbb{B}_\lambda(\xi) \cdot \mathbb{A}(\xi) \widehat{\varphi r_j}(\xi)}{1 + |\mathbb{B}_\lambda(\xi)|^2} + \frac{\mathbb{B}_\lambda(\xi) \cdot \mathbb{A}(\xi) \widehat{w_j}(\xi)}{1 + |\mathbb{B}_\lambda(\xi)|^2} + \frac{\widehat{\varphi\sigma_j}(\xi)}{1 + |\mathbb{B}_\lambda(\xi)|^2}.$$

Then, we express

$$\varphi\sigma_j = f_j + g_j.$$

We see that

$$f_j = \mathcal{F}^{-1}(m_0(\xi)\mathcal{F}(r_j)), \quad g_j = \mathcal{F}(m_1(\xi)\mathcal{F}(w_j)),$$

where we choose w_j by w_j and r_j . With almost the same arguments in the proof of Theorem 2.11(2), $\{g_j\}$ is precompact in L^1 . By (3.2) and Theorem A.4, for any $\lambda > 0$,

$$\lambda \mathcal{L}^d(\{x : |f_j| > \lambda\}) \leq C|r_j|(B_1) \rightarrow 0$$

as $j \rightarrow +\infty$. Note that $\varphi\nu_j \geq 0$, we see that $0 \leq f_j^{-1} \leq |g_j|$. By the precompactness of g_j , we see that $f_j^{-1} \rightarrow 0$ in L^1 . Moreover, by the definition of f_j , for any $\psi \in C_0^\infty$, it follows from Parseval's identity that

$$\int f_j \psi = \int r_j \tilde{\psi} \rightarrow 0$$

as $j \rightarrow +\infty$ for some $\tilde{\psi}$ depending only on ψ . Since ψ is arbitrary and $f_j^{-1} \rightarrow 0$ in L^1 , we complete the proof. $\square$

Remark 3.9. In the proof above, we abuse the notation of $\mathcal{M}(\Omega)$ and L^1 . It may be safer to analyze $B_{\frac{1}{2}}$ than on B_1 .

3.3. Dimension estimates for $\mathcal{A}$ -free measures. In this subsection, we will not provide proof for most of the results but only show their relations to give a general picture of this field. Let us first consider some motivations.

Recall that we can express the equivalent form of Theorem 3.6 as follows.

Theorem 3.10. *If a set $E \subset \mathbb{R}^d$ satisfies that $\mathcal{L}^d(E) = 0$ and $|\mu|(E) > 0$, then*

$$\frac{d\mu}{d|\mu|} \in \Lambda_{\mathcal{A}} \quad \text{for } |\mu|\text{-a.e. } x \in E.$$

A natural question is if we can replace $\mathcal{L}^d = \mathcal{H}^d$ in the above theorem to $\mathcal{H}^\ell$ or more general measures. Let us now consider more examples.

Let $\mu = \lambda\delta_0$ and $\mathcal{A}$ be a differential operator with order k . Assume that $\mathcal{A}(\lambda\delta_0) = 0$. Then,

$$\mathbb{A}(\xi)\lambda\widehat{\delta_0}(\xi) = 0.$$

Since $\widehat{\delta_0} = 1$, it follows from the arbitrariness of ξ that

$$\lambda \in \bigcap_{|\xi|=1} \mathbb{A}(\xi).$$

Define $\sigma := \mathcal{H}^k \llcorner \pi$, where $\pi \in \mathbb{G}(d, k)$ ($\mathbb{G}(d, k)$ is the Grassmannian manifold). If $\mathcal{A}(\lambda\sigma) = 0$, we have

$$\mathbb{A}(\xi)\lambda\mathcal{H}^{d-k} \llcorner \pi^\perp(\xi) = 0.$$

As a result,

$$\lambda \in \bigcap_{|\xi|=1, \xi \in \pi^\perp} \ker \mathbb{A}(\xi).$$

Through the above examples, we present the following definition.

Definition 3.11 (ℓ -wave cone). For $\ell \in \mathbb{Z} \cap [1, d]$, we define the ℓ -dimensional wave cone as

$$\Lambda_{\mathcal{A}}^\ell := \bigcap_{\pi \in \mathbb{G}(d, \ell)} \bigcup_{\xi \in \pi \setminus \{0\}} \ker \mathbb{A}(\xi).$$

It is easy to see that

$$\lambda \notin \Lambda_{\mathcal{A}}^\ell \iff (\mathcal{A} \llcorner \pi)\lambda \text{ is elliptic for some } \pi \in \mathbb{G}(d, \ell),$$

where we define $\mathcal{A} \llcorner \pi$ as

$$C^\infty(\pi, \mathbb{R}^m) \ni \varphi \mapsto (\mathcal{A} \llcorner \pi)(\varphi) := \mathcal{A}(\varphi \circ p_\pi)$$

with p_π the orthogonal projection onto π . It directly follows from Definition 3.11 that

$$\Lambda_{\mathcal{A}}^1 = \bigcap_{\xi \in \mathbb{R}^d \setminus \{0\}} \ker \mathbb{A}(\xi) \subset \Lambda_{\mathcal{A}}^j \subset \Lambda_{\mathcal{A}}^\ell \subset \Lambda_{\mathcal{A}}^d = \Lambda_{\mathcal{A}}, \quad 1 \leq j \leq \ell \leq d.$$

With the above definition given in hand, we have the following theorem.

Theorem 3.12 (Arroyo-De Philippis-Hirsch-Rindler, [2]). *Let $\Omega \subset \mathbb{R}^d$ be open, and $\mathcal{A}$ be a differential operator of order k . Assume that $\mu \in \mathcal{M}(\Omega, \mathbb{R}^m)$ is an $\mathcal{A}$ -free measure on Ω . If $E \subset \Omega$ is a Borel set with $\mathcal{H}^\ell(E) = 0$ for some $\ell \in \mathbb{Z} \cap [0, d]$, then*

$$\frac{d\mu}{d|\mu|}(x) \in \Lambda_{\mathcal{A}}^\ell \quad \text{for } |\mu|\text{-a.e. } x \in E.$$

Remark 3.13. We can replace $\mathcal{H}^\ell$ in the above theorem by integral geometric measure $\mathcal{I}^\ell$. We define it as

$$\mathcal{I}^\ell(E) := \int_{\mathbb{G}(d, \ell)} \int_{\pi} \mathcal{H}^0(E \cap p_\pi^{-1}(x)) d\mathcal{H}^\ell(x) d\gamma_{\ell, d}(\pi),$$

where $\gamma_{\ell, d}$ is the unique $\mathbb{O}(d)$ -invariant probability measure on $\mathbb{G}(d, \ell)$.

We now recall the Federer-Besicovitch characterization of rectifiability. One can refer to [11, Section 3] for more information.

Theorem 3.14 (Federer-Besicovitch). *Let $E \subset \mathbb{R}^d$ be a subset with $0 < \mathcal{H}^\ell(E) < +\infty$. Then, $\mathcal{I}^\ell(E) = 0$ if and only if E is purely unrectifiable, that is, for any ℓ -dimensional manifold Σ , $\mathcal{H}^\ell(E \cap \Sigma) = 0$.*

Combining this and Theorem 3.12, we have the following corollaries.

Corollary 3.15. *Assume that μ is a Radon measure such that the following properties hold.*

- (1) $\mathcal{H}^\ell \ll |\mu|$ (μ is at most ℓ -dimensional.)
- (2) $|\mu| \ll \mathcal{I}^\ell$.

Then, μ is ℓ -rectifiable.

Corollary 3.16. *Assume that μ is $\mathcal{A}$ -free, and there is $\ell \in \mathbb{Z} \cap [1, d]$ such that*

$$\frac{d\mu}{d|\mu|}(x) \notin \Lambda_{\mathcal{A}}^\ell \quad \text{for } |\mu|\text{-a.e. } x.$$

Then

$$|\mu| \ll \left\{ x : \liminf_{R \rightarrow 0^+} \frac{|\mu|(B_R(x))}{R^\ell} > 0 \right\}$$

is ℓ -rectifiable.

Next, we will use Corollary 3.16 to obtain some rectifiability results in geometric measure theory.

Example 3.17. For $\mathcal{A} = \text{curl}$, it follows from Example 2.10 that $\dim \ker \mathbb{A}(\xi) = 1$ for any $\xi \neq 0$. We see that $\Lambda_{\mathcal{A}}^{d-1} = \{0\}$. Applying Corollary 3.16, we obtain the $(d-1)$ -rectifiability of BV functions (see [11, Section 3.4]).

Example 3.18. For $\mathcal{A} = \text{div}$, we see that

$$\mathcal{A}_{\text{div}}^\ell = \{M : \text{rank } M \leq \ell - 1\}.$$

Consider u_j in Theorem 1.18, we see that

$$\text{div}(|\nabla u_j| \text{Id} - 2\nabla u_j \otimes \nabla u_j) = 0.$$

Define

$$T_j := |\nabla u_j| \text{Id} - 2\nabla u_j \otimes \nabla u_j.$$

We see that

$$T_j \rightharpoonup^* T, \quad \mu_j := |\nabla u_j|^2 dx \rightharpoonup^* |\nabla u|^2 dx + \nu,$$

in the sense of Radon measures. Then

$$T = P\mu, \quad P = \int (I - 2y \otimes y) d\nu_x(y)$$

with $\text{rank } P \geq d - 2$. As a result, Theorem 1.18 follows from Corollary 3.16.

APPENDIX A. SOME RESULTS IN FOURIER ANALYSIS

A.1. Characterization of precompact sets in $L^p(\mathbb{R}^n)$.

Theorem A.1 (Kolmogorov-Riesz). *Let $p \in (1, +\infty)$. A subset K of $L^p(\mathbb{R}^n)$ precompact, if and only if the following properties hold.*

- (1) K is bounded in $L^p(\mathbb{R}^n)$.
- (2) For any $\varepsilon > 0$, there is $R > 0$ such that, for any $f \in K$,

$$\int_{|x| > R} |f|^p dx < \varepsilon^p.$$

- (3) For any $\varepsilon > 0$ there is $\rho > 0$ such that, for any $f \in K$ and $y \in \mathbb{R}^n$ with $|y| < \rho$,

$$\int_{\mathbb{R}^n} |f(x+y) - f(x)|^p dx < \varepsilon^p.$$

Remark A.2. The three properties refer to the boundedness, equitight, and equicontinuity for the translation, respectively.

For the proof of Theorem A.1, we recommend readers the [7]. Consequently, we have the following property when $p = 2$.

Corollary A.3. *A subset K of $L^2(\mathbb{R}^n)$ precompact, if and only if the following properties hold.*

- (1) *K is bounded in $L^2(\mathbb{R}^n)$.*
- (2) *For any $\varepsilon > 0$, there is $R > 0$ such that, for any $f \in K$,*

$$\int_{|x|>R} |f|^2 dx < \varepsilon^2.$$

- (3) *For any $\varepsilon > 0$, there is $R > 0$ such that, for any $f \in K$,*

$$\int_{|x|>R} |\widehat{f}|^2 dx < \varepsilon^2.$$

Proof. Assume that $K \subset L^2(\mathbb{R}^n)$ is precompact, we claim that

$$\widehat{K} := \{\widehat{f} : f \in K\}$$

is also a precompact. It is because $\{\widehat{f}_j\}$ is a Cauchy sequence with respect to the norm of L^2 if and only if $\{f_j\}$ is a Cauchy sequence with respect to the same norm. As a result, we can verify the third property by the second result in Theorem A.1. On the other hand, Consider the translation difference

$$\|f(\cdot + h) - f\|_{L^2}^2 = \int_{\mathbb{R}^n} |f(x+h) - f(x)|^2 dx.$$

By Parseval's theorem, we have

$$\|f(\cdot + h) - f\|_{L^2}^2 = \|\mathcal{F}(f(\cdot + h) - f)\|_{L^2}^2 = \int_{\mathbb{R}^n} |\exp(2\pi i \xi \cdot h) - 1|^2 |\widehat{f}(\xi)|^2 d\xi.$$

Split the integral over $|\xi| \leq S$ and $|\xi| > S$. Choose $S > 0$ such that

$$\int_{|\xi|>S} |\widehat{f}(\xi)|^2 d\xi < \frac{\varepsilon^2}{8}$$

for any $f \in K$, which is possible by uniform frequency decay. Since $|\exp(2\pi i \xi \cdot h) - 1| \leq 2$, it follows that

$$\int_{|\xi|>S} |\exp(2\pi i \xi \cdot h) - 1|^2 |\widehat{f}(\xi)|^2 d\xi \leq 4 \int_{|\xi|>S} |\widehat{f}(\xi)|^2 d\xi < 4 \cdot \frac{\varepsilon^2}{8} = \frac{\varepsilon^2}{2}. \quad (\text{A.1})$$

Also, we note that

$$|\exp(2\pi i \xi \cdot h) - 1| \leq 2\pi |\xi| |h| \leq 2\pi S |h|$$

when $|\xi| \leq S$. As a result,

$$\int_{|\xi|\leq S} |\exp(2\pi i \xi \cdot h) - 1|^2 |\widehat{f}(\xi)|^2 d\xi \leq (2\pi S |h|)^2 \int_{|\xi|\leq S} |\widehat{f}(\xi)|^2 d\xi \leq (2\pi S |h|)^2 M^2,$$

where we assume that

$$\sup_{f \in K} \|f\|_{L^2} \leq M.$$

Choose $\delta = \frac{\varepsilon}{2\pi S M \sqrt{2}}$. If $|h| < \delta$, then

$$(2\pi S |h|)^2 M^2 < \left(2\pi S \cdot \frac{\varepsilon}{2\pi S M \sqrt{2}}\right)^2 M^2 = \left(\frac{\varepsilon}{M \sqrt{2}}\right)^2 M^2 = \frac{\varepsilon^2}{2}.$$

Combining (A.1), we have

$$\|f(\cdot + h) - f\|_{L^2}^2 < \frac{\varepsilon^2}{2} + \frac{\varepsilon^2}{2} = \varepsilon^2$$

for all $f \in K$ when $|h| < \delta$. Then, the result follows from Theorem A.1. $\square$

A.2. The Mihlin-Hörmander multiplier theorem.

Theorem A.4. *Let $m(\xi)$ be a complex-valued bounded function on $\mathbb{R}^d \setminus \{0\}$ that satisfies*

$$|\partial_\xi^\alpha m(\xi)| \leq C_\alpha |\xi|^{-|\alpha|}$$

for any multi-index $\alpha = (\alpha_1, \dots, \alpha_d)$, then the operator

$$f \mapsto \mathcal{F}^{-1}(m\mathcal{F}(f))$$

is weak $(1, 1)$ and strong (p, p) for any $p \in (1, +\infty)$.

Proof. It is a direct consequence of [6, Theorem 6.2.7]. □

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